

SWOT in the Surfzone

Applications, Limitations, and Insights for SWOT in Wave Dominated Nearshore Environments

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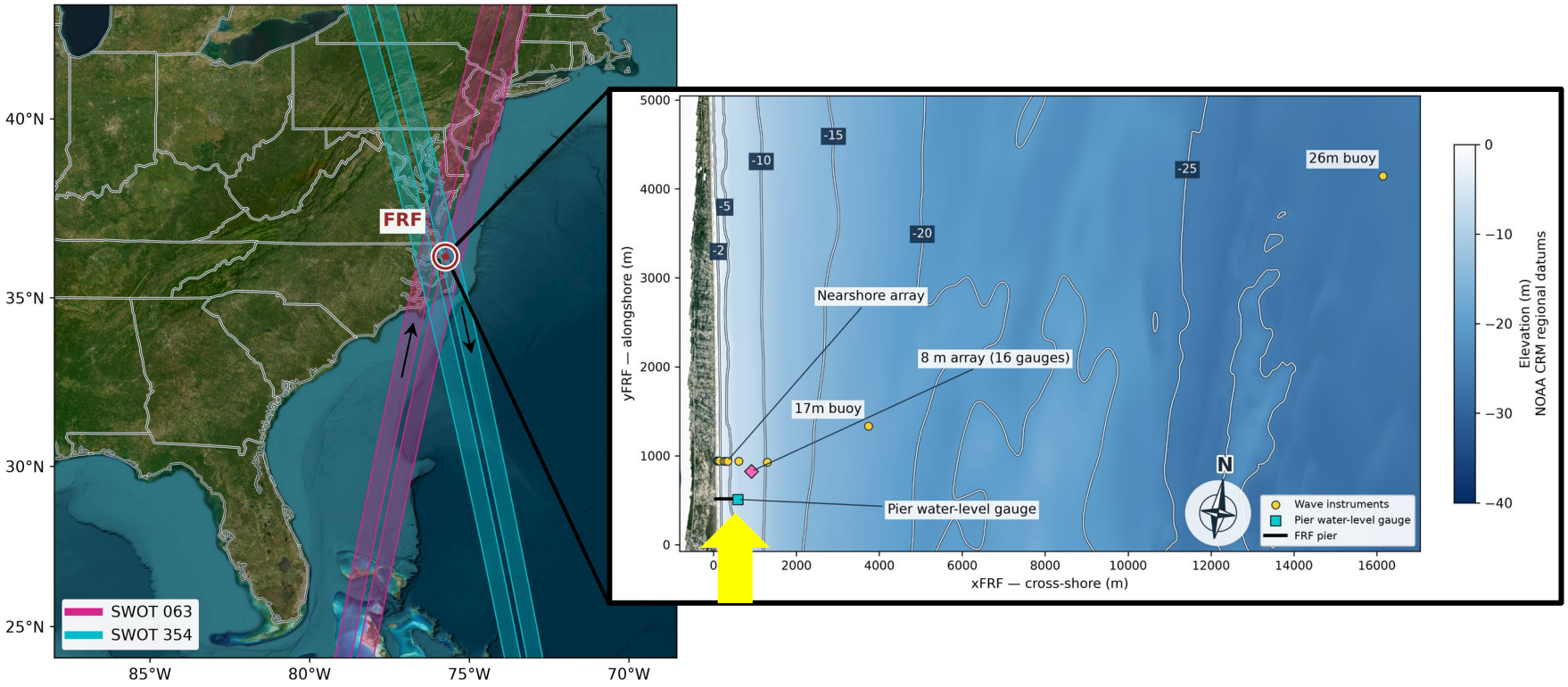
IGE

Topics

- Water surface elevation in the nearshore and inner-surfzone
- Wave setup with HR pixel cloud
- Wave height with LR Unsmoothed (250 m)

Nearshore Water Surface Elevation (WSE)

Study Site

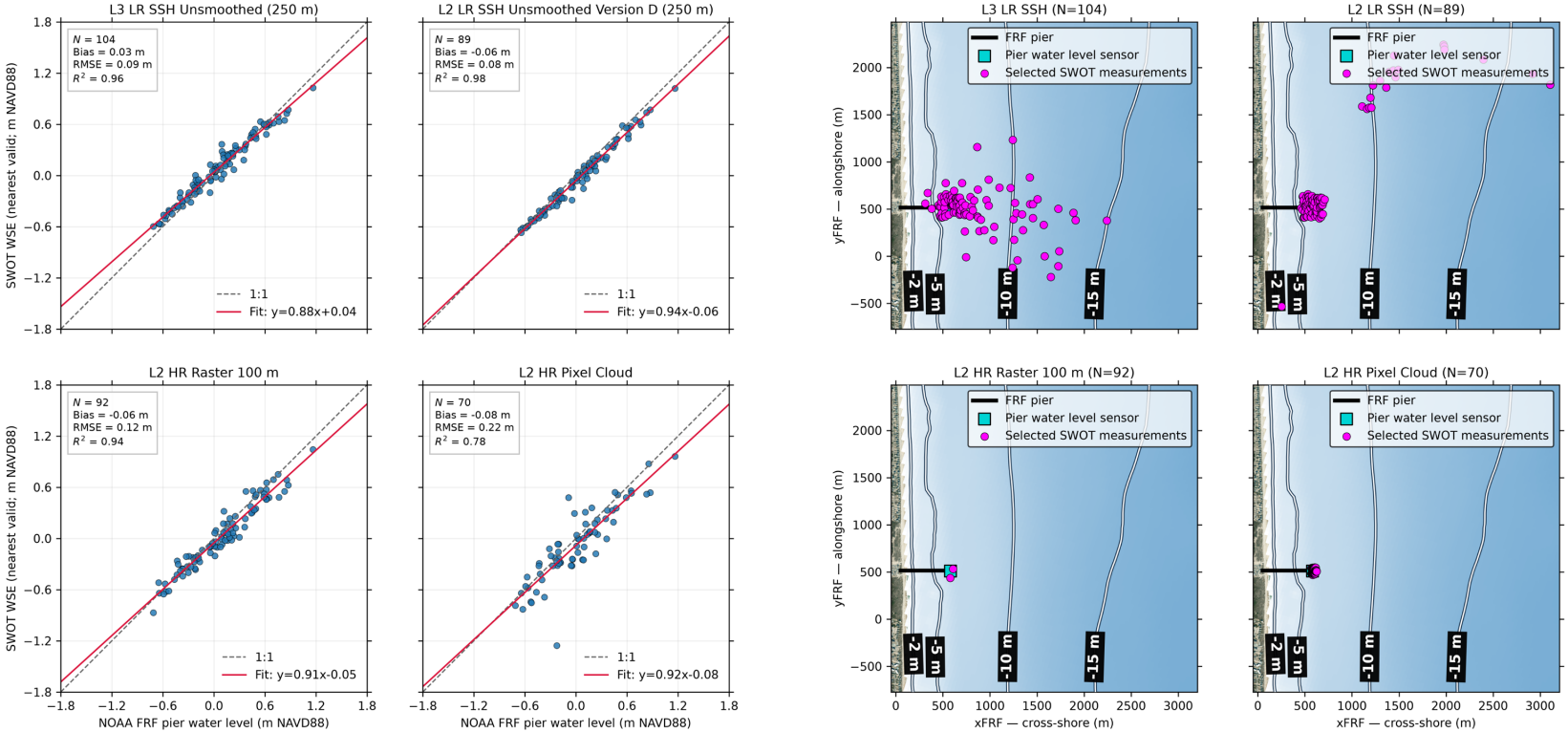

40°N
35°N
30°N
25°N
85°W
80°W
75°W
70°W
FRF
SWOT 063
SWOT 354
yFRF — alongshore (m)
xFRF — cross-shore (m)
Nearshore array
8 m array (16 gauges)
17m buoy
26m buoy
Pier water-level gauge
N
Wave instruments
Pier water-level gauge
FRF pier
Elevation (m)
NOAA CRM regional datums

Nearshore Water Surface Elevation (WSE)

Products:

- L3 LR SSH Unsmoothed (250 m)
- L2 LR SSH Unsmoothed (250 m)
- L2 HR Raster (100 m)
- L2 HR Water Mask Pixel Cloud (22 x 25/45 m)

Nearshore Water Surface Elevation (WSE)



WSE in the inner-surfzone

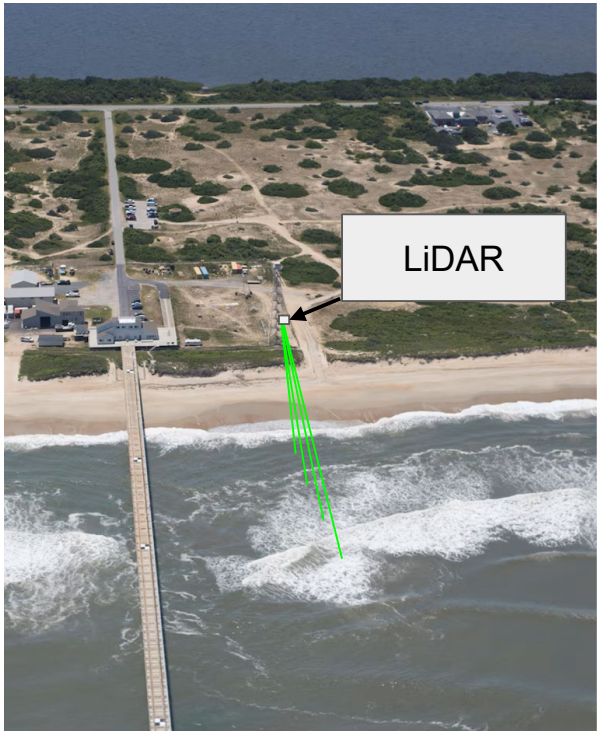


Illustration only, not actual configuration

WSE in the inner-surfzone

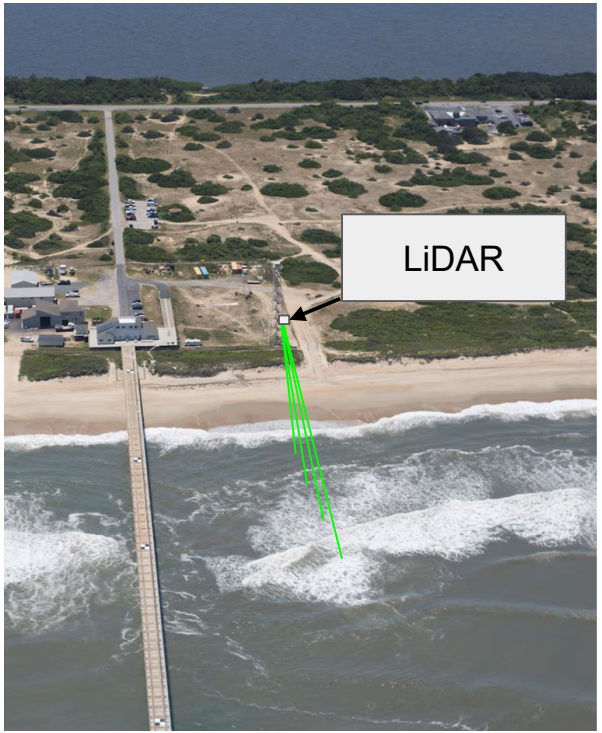
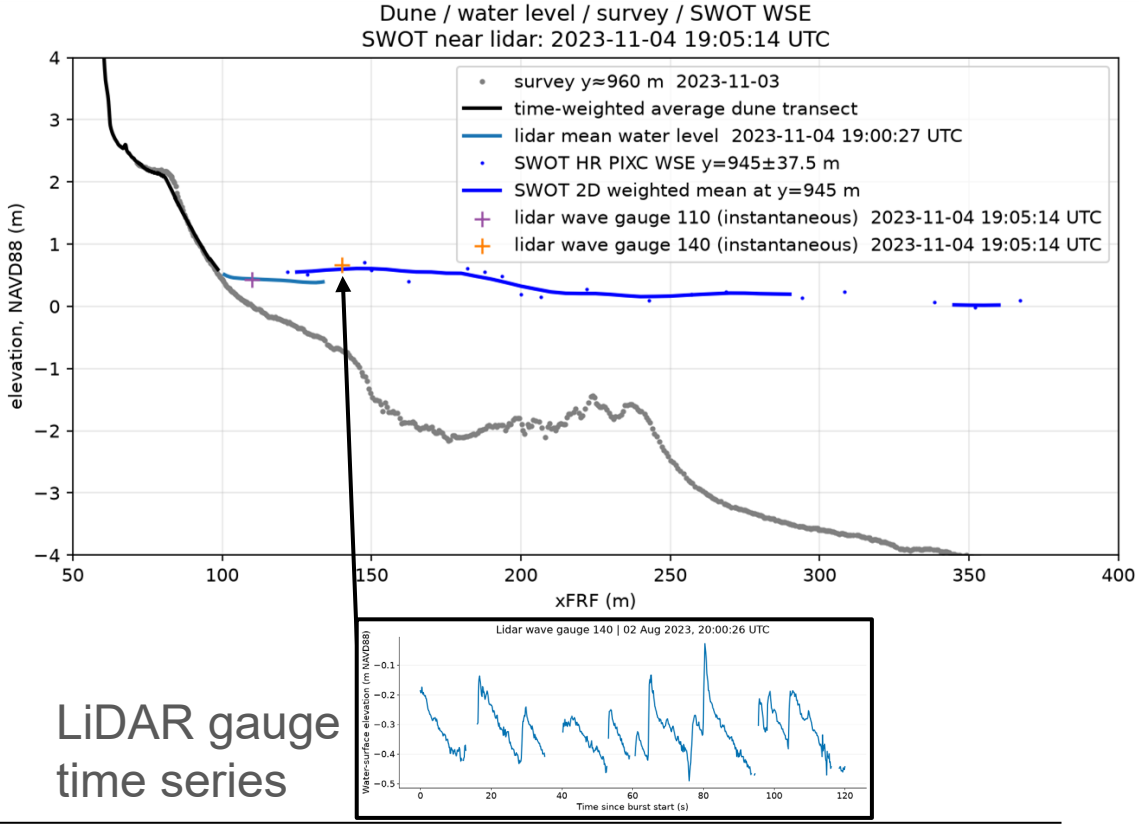
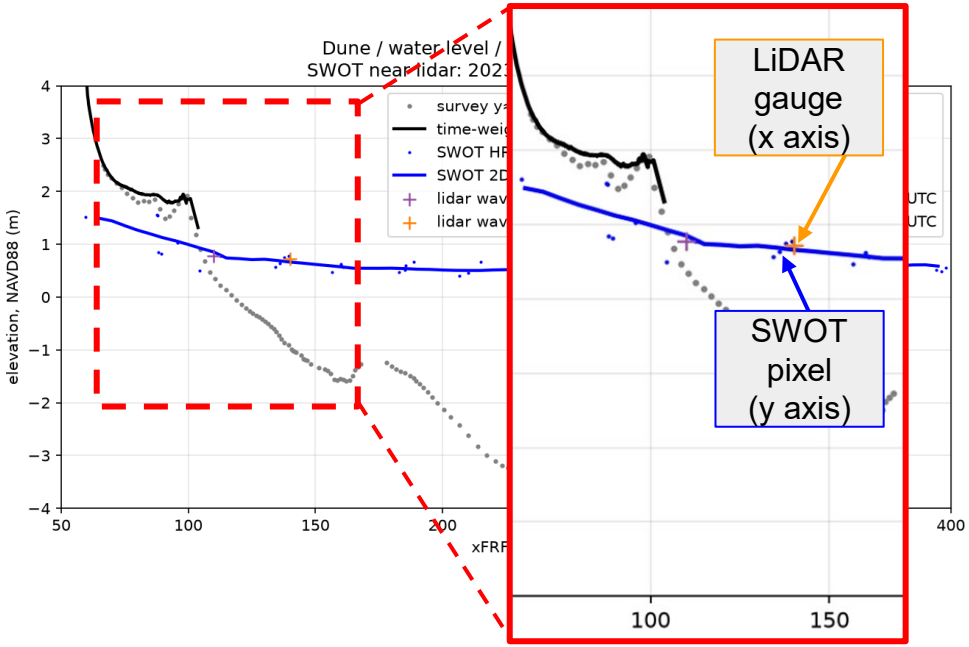


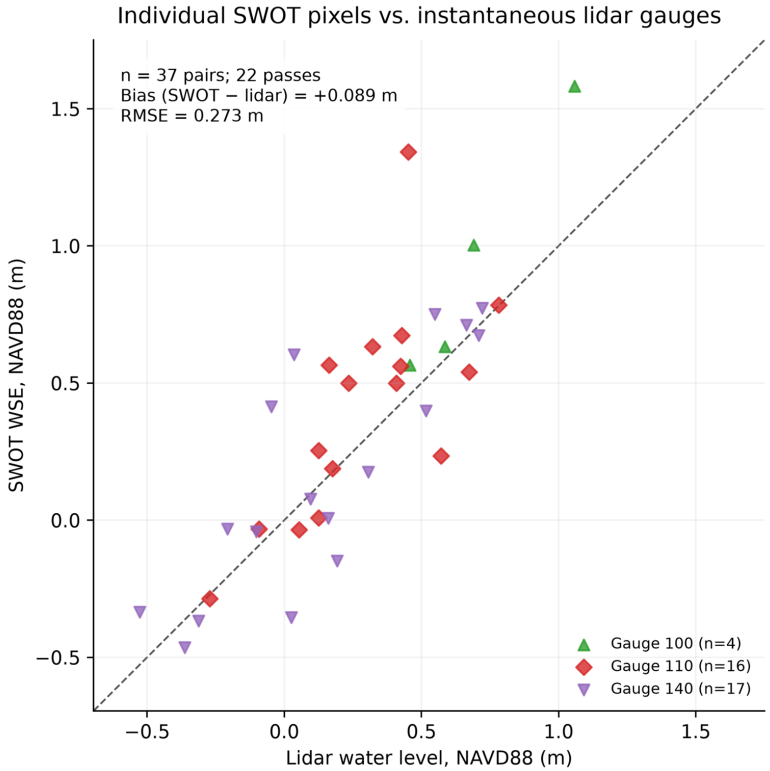
Illustration only, not actual configuration



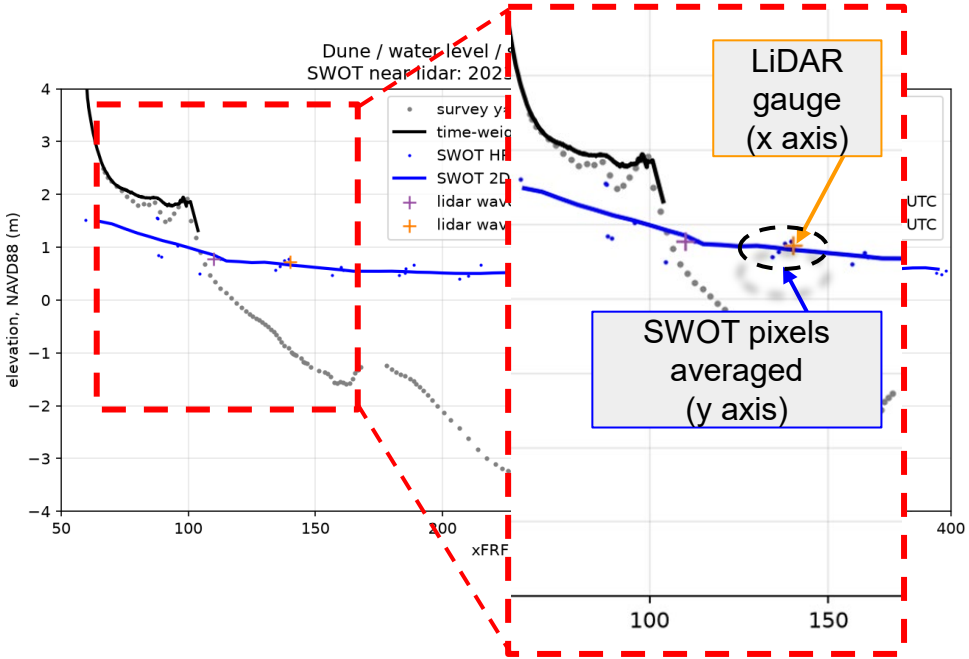
LiDAR gauge time series

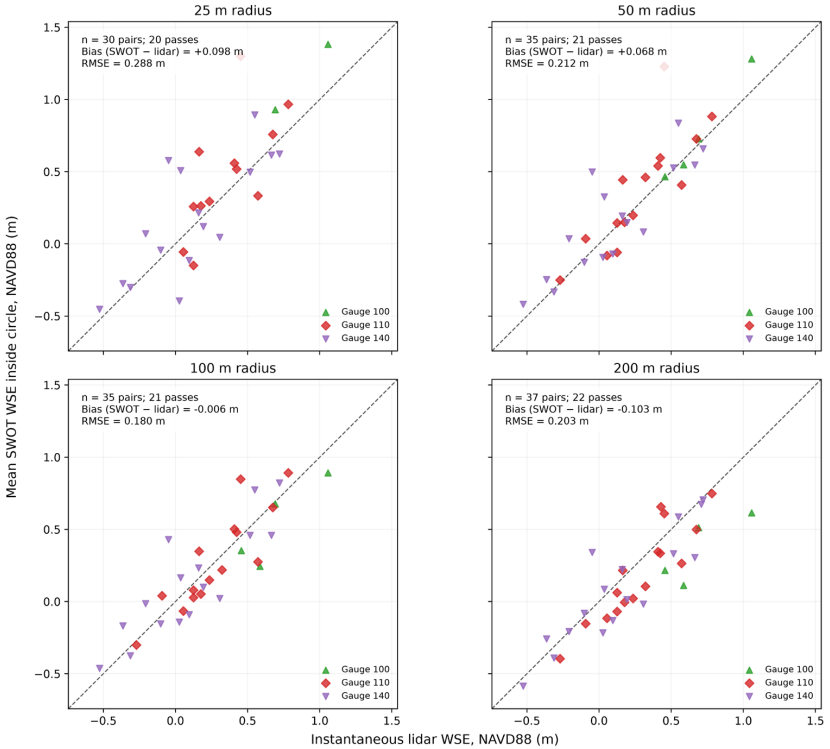
WSE in the inner-surfzone- SWOT single pixel





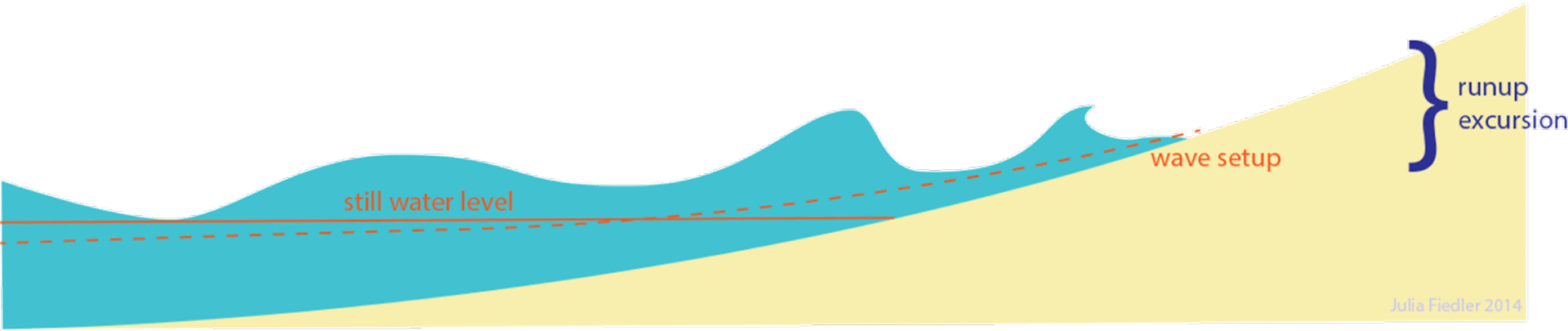
WSE in the inner-surfzone- SWOT spatially averaged





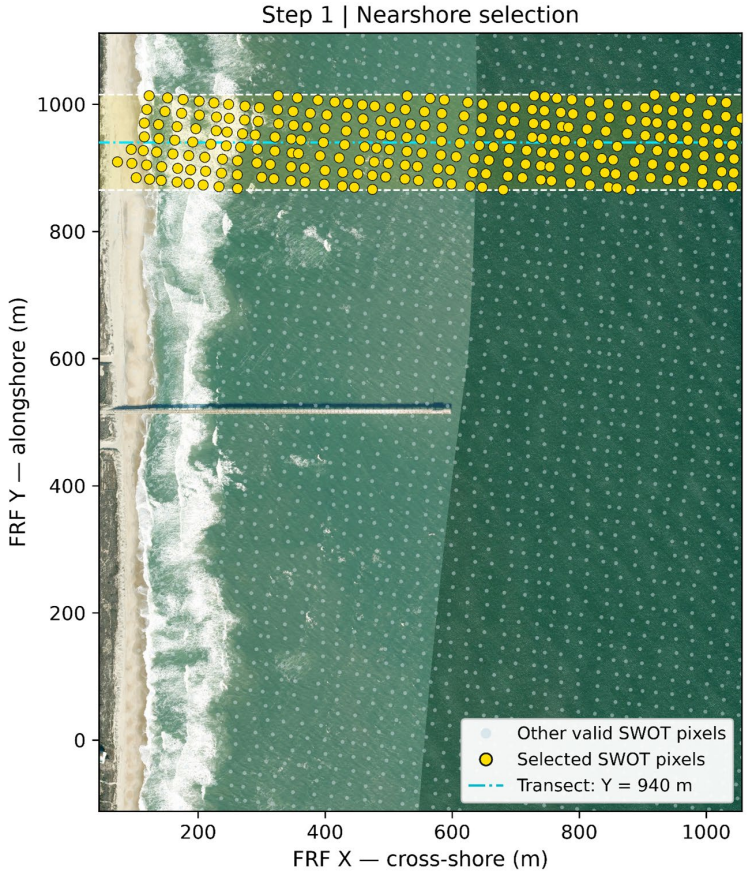
Wave setup with HR pixel cloud

SWOT Wave Setup

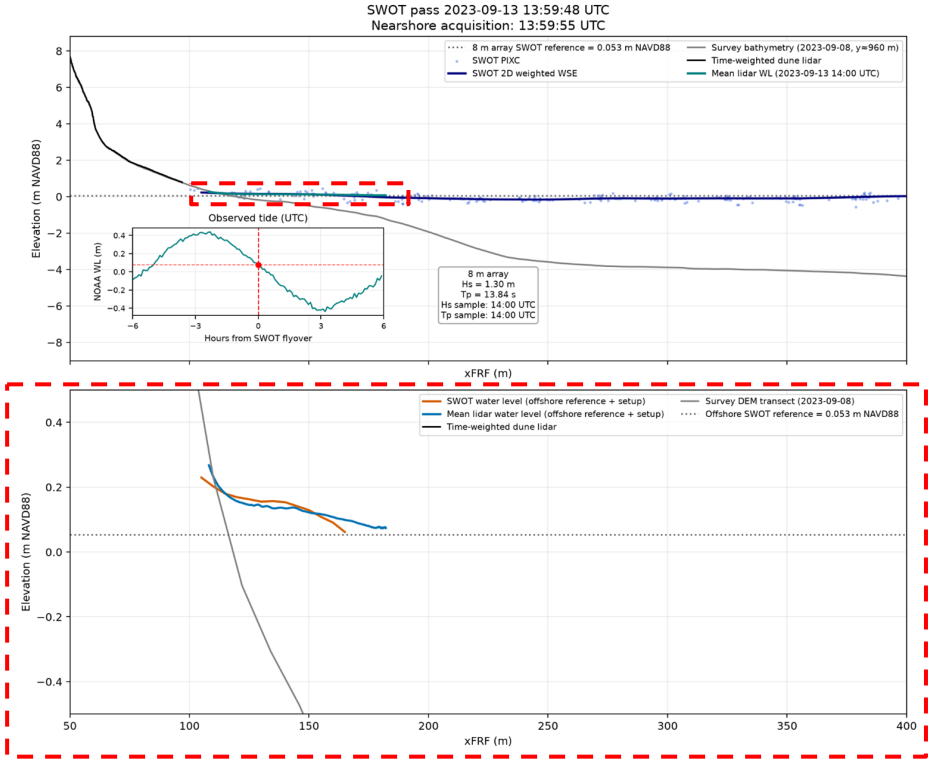


SWOT Wave Setup

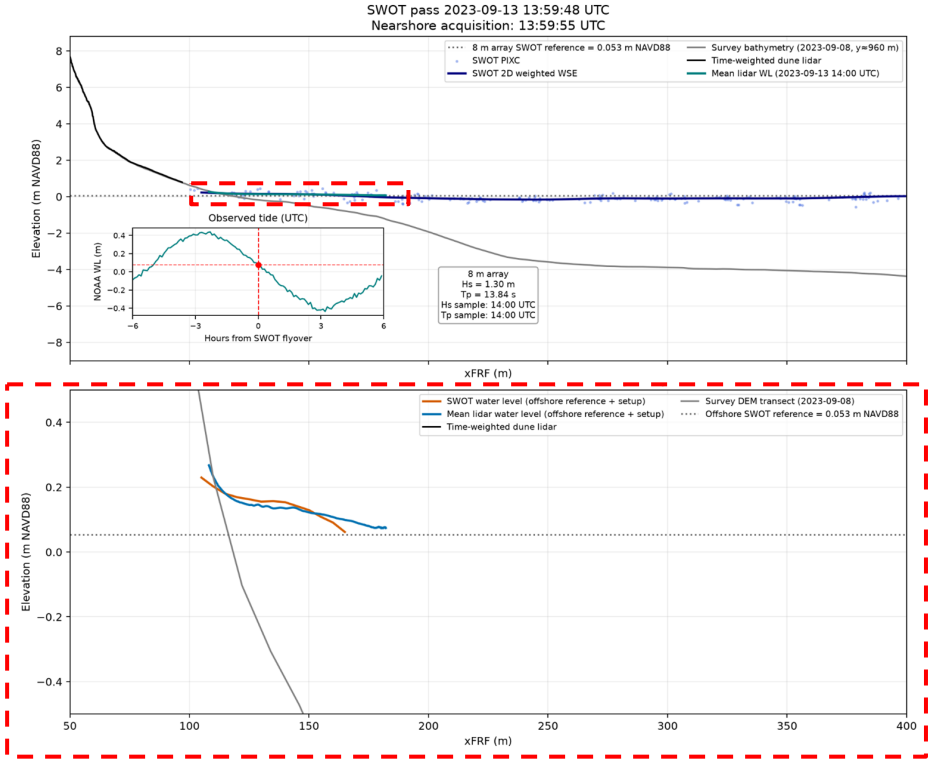
- **SWOT transect:** Averaged pixels within ±75 m of the y = 940 m transect

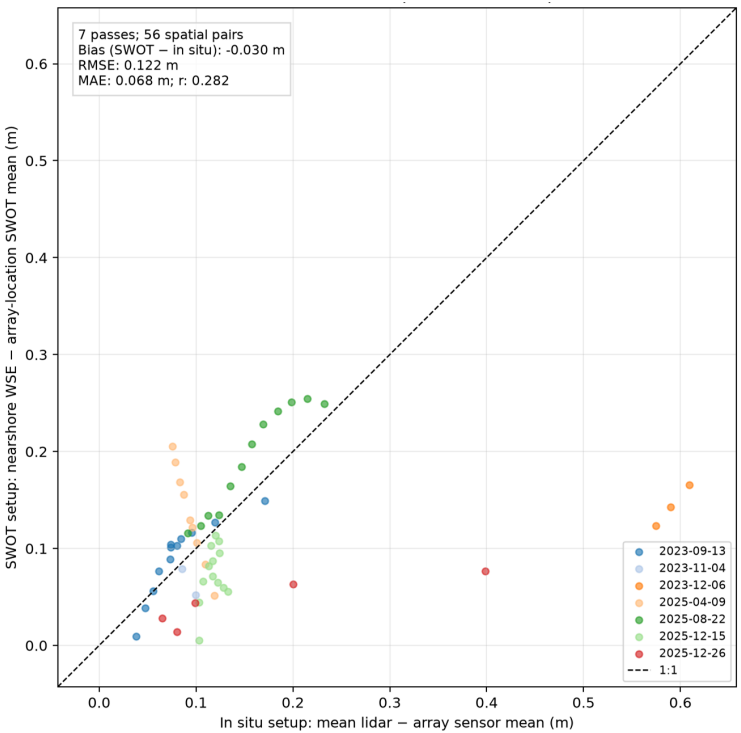


SWOT Wave Setup- Time averaged in-situ

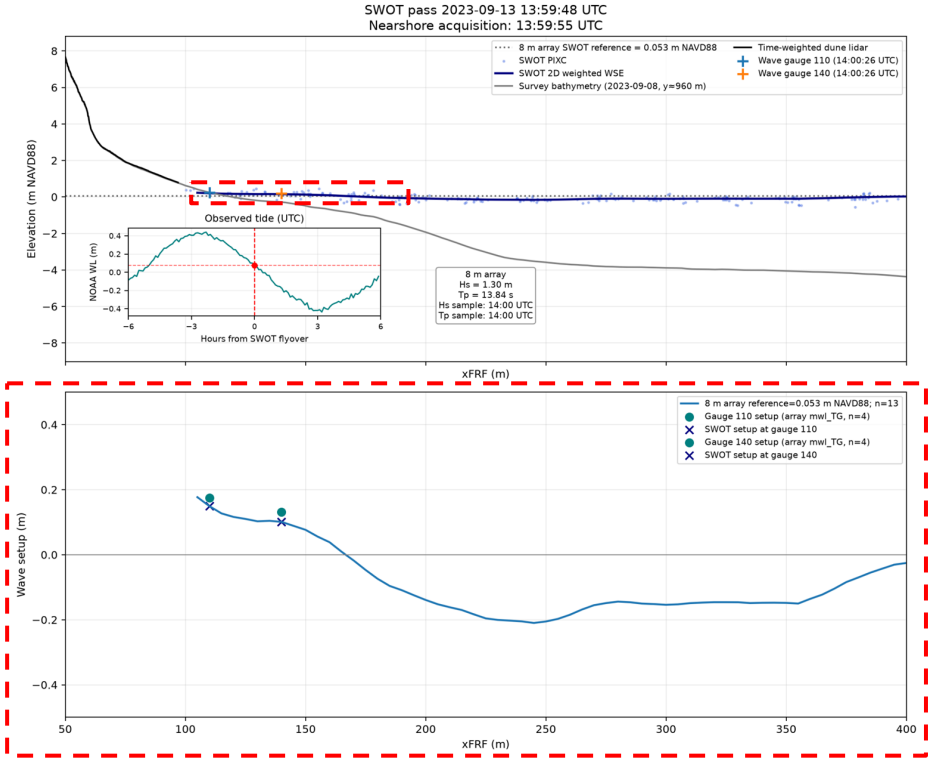


SWOT Wave Setup- Time averaged in-situ

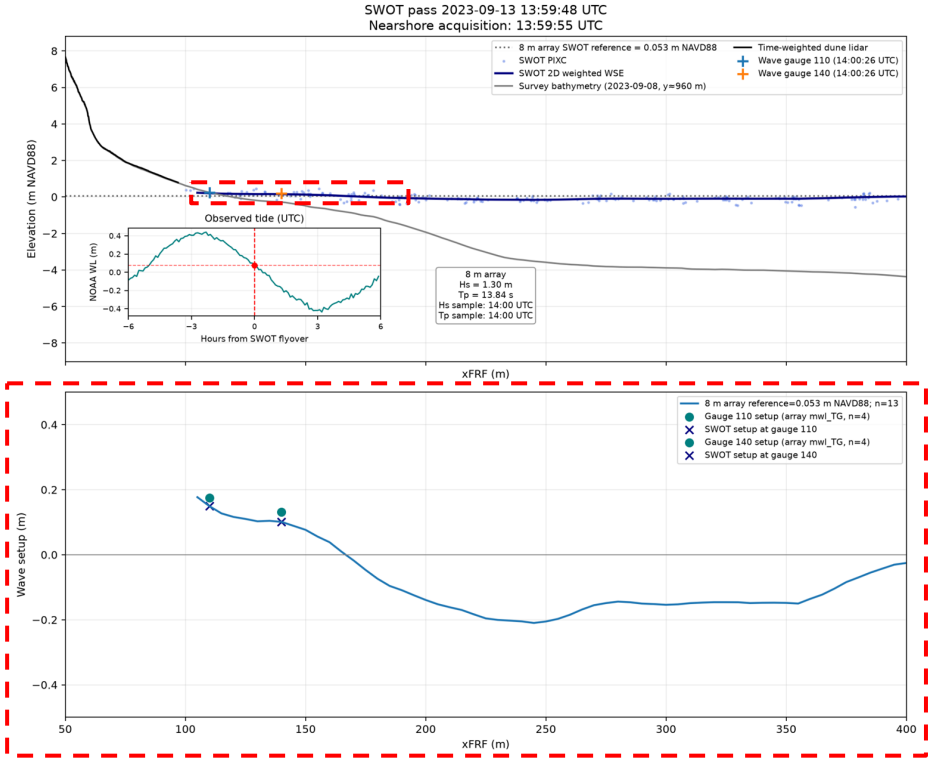


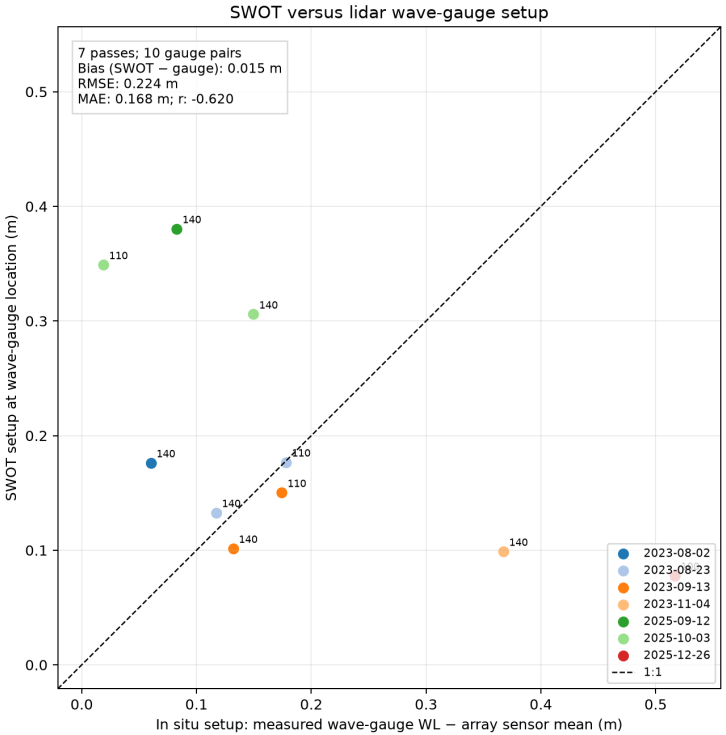


SWOT Wave Setup- Instantaneous in-situ



SWOT Wave Setup- Instantaneous in-situ





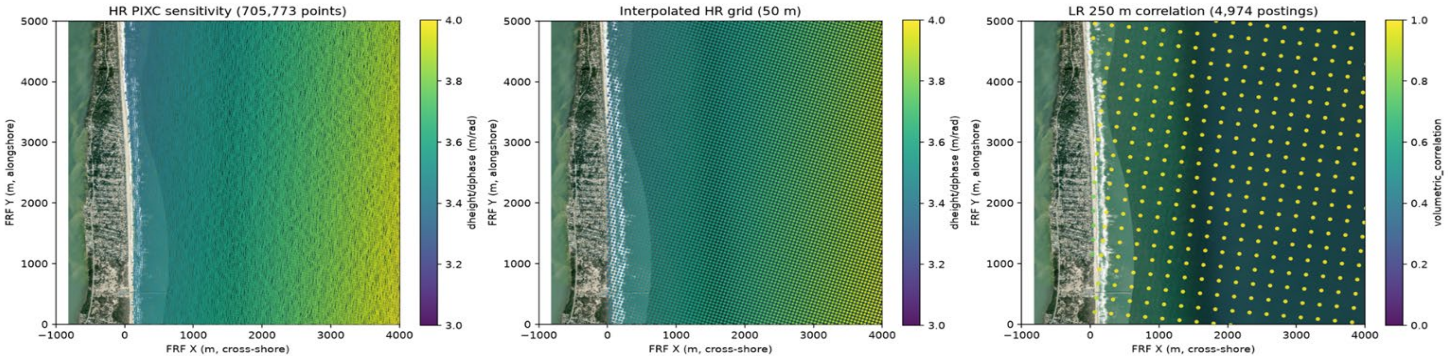
Significant wave height using LR Unsmoothed

Significant wave height using LR Unsmoothed

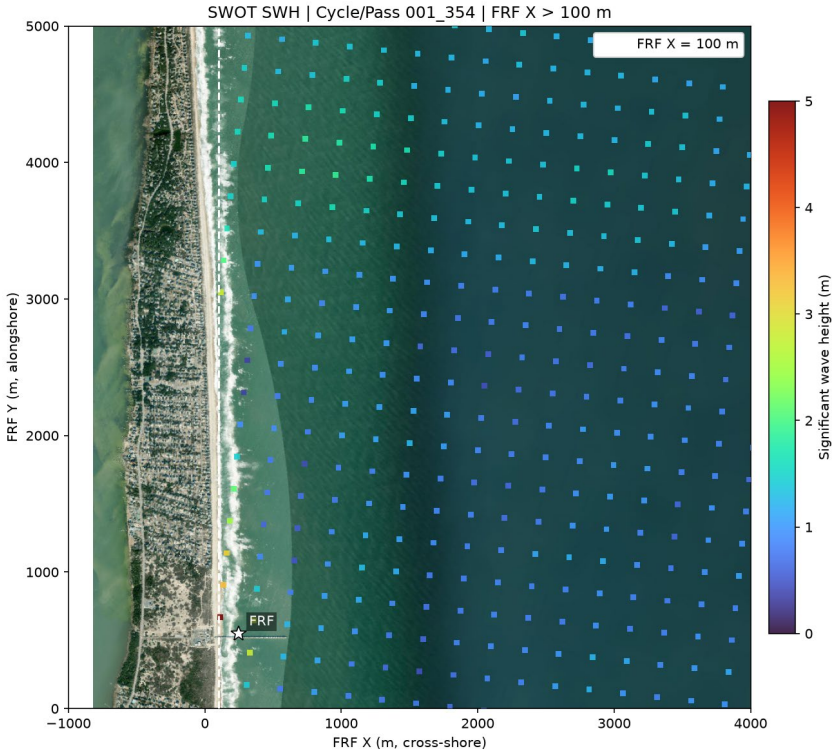
$$H_s = 4\frac{\partial \phi}{\partial h}\sqrt{-2\ln\gamma}$$

Height sensitivity to phase
Dheight_dphase in the **HR pixel cloud** product

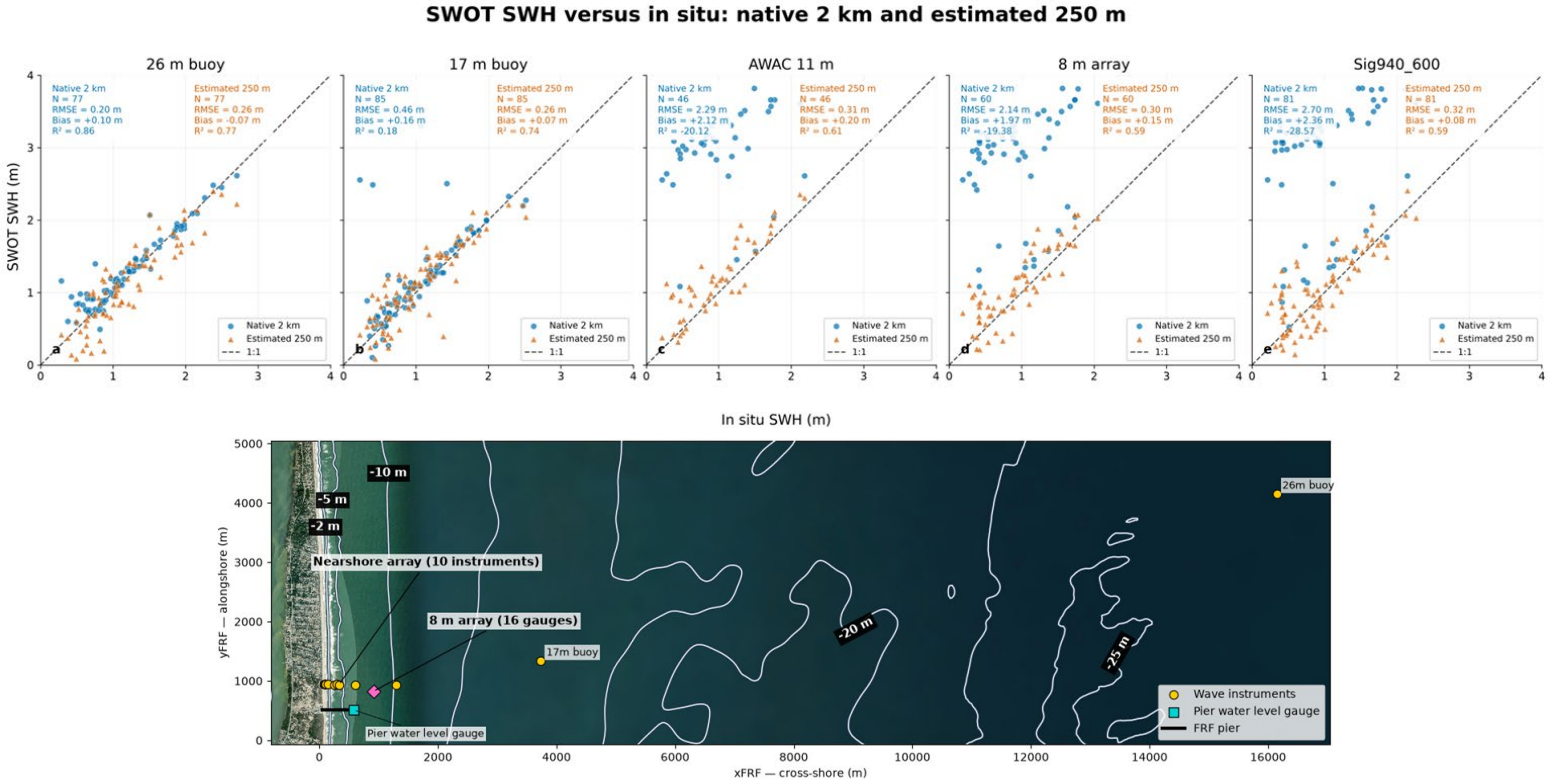
Volumetric Correlation
volumetric_correlation in the **LR Unsmoothed**



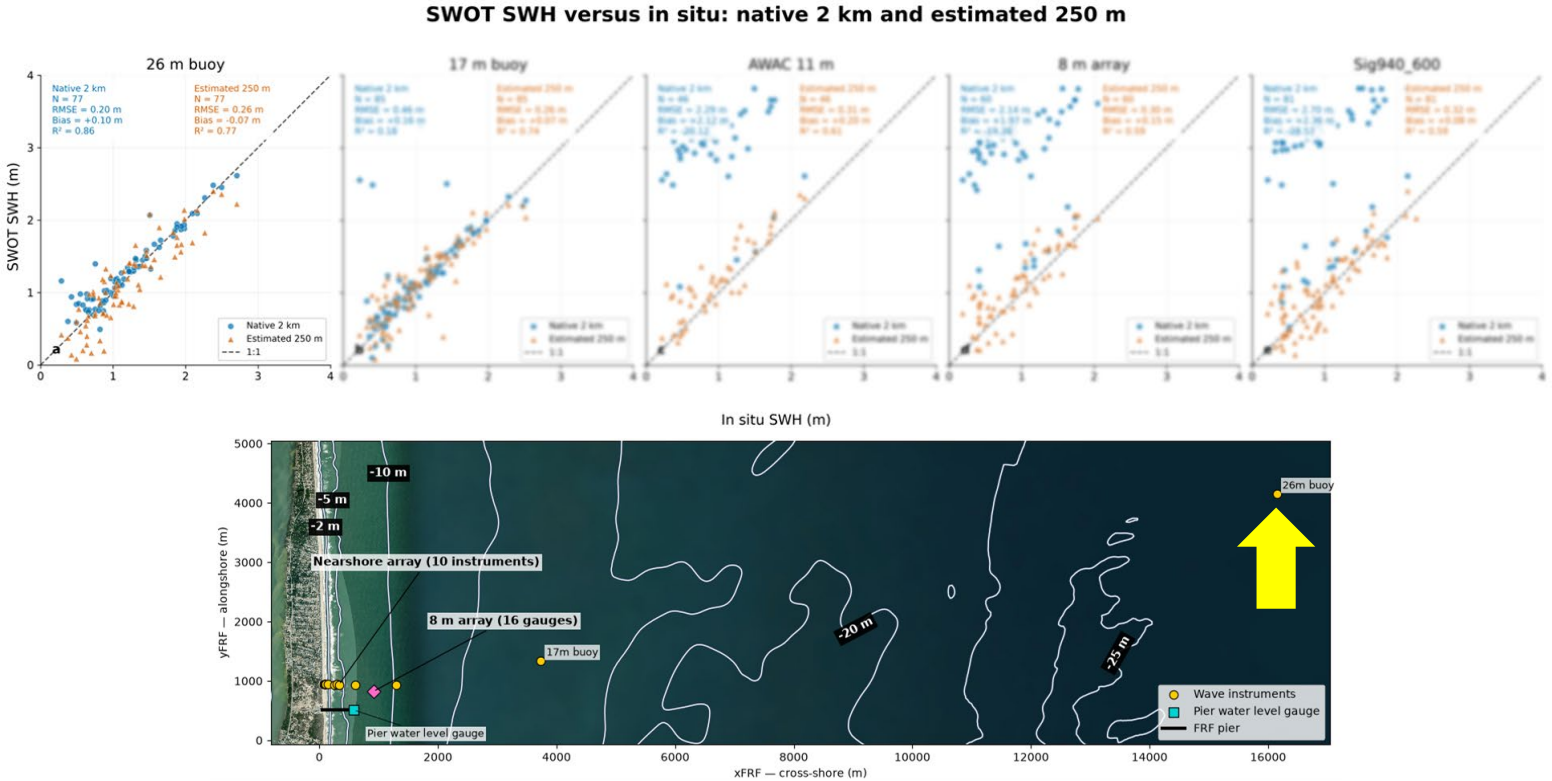
Significant wave height using LR Unsmoothed



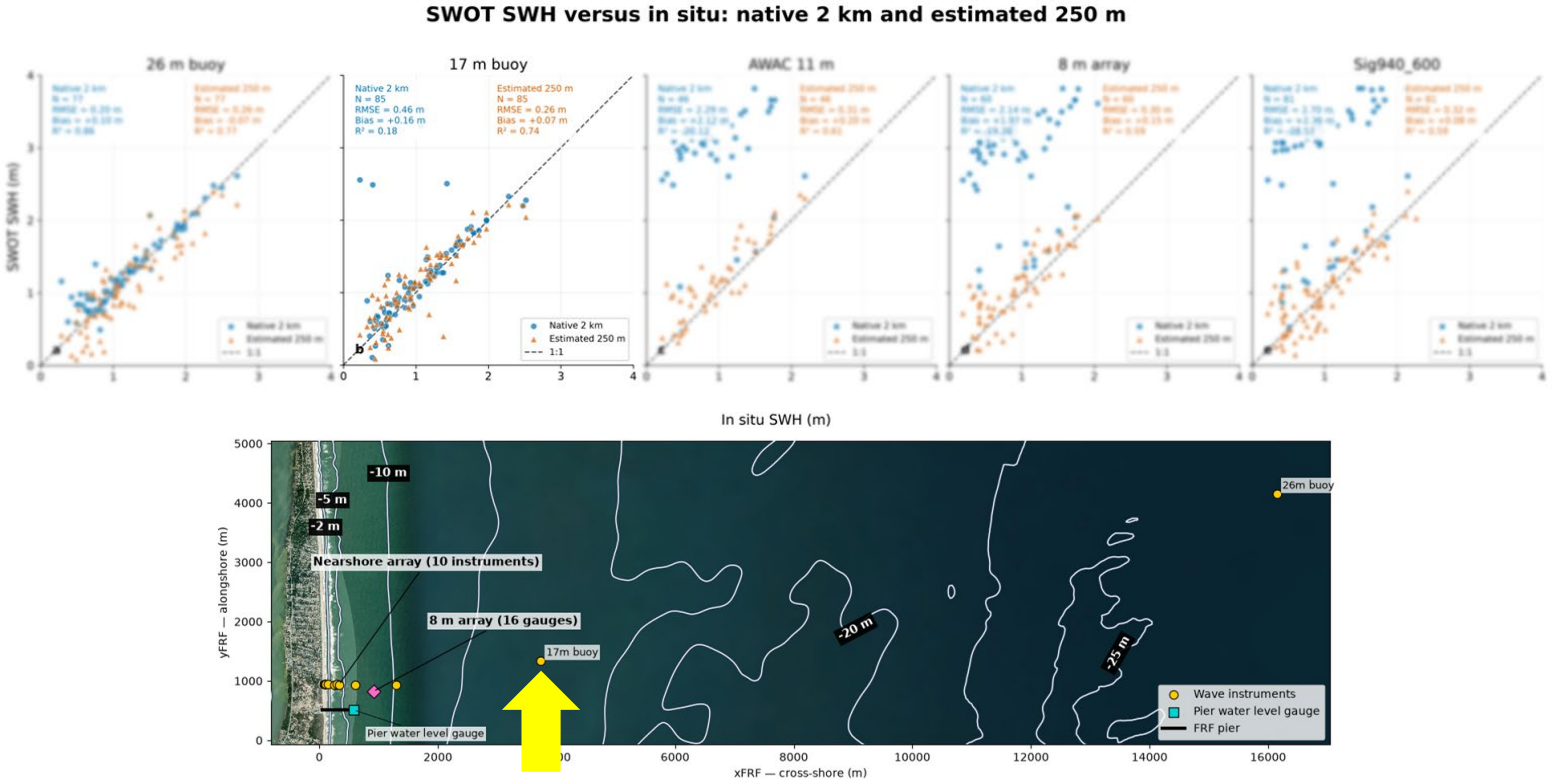
Significant wave height using LR Unsmoothed



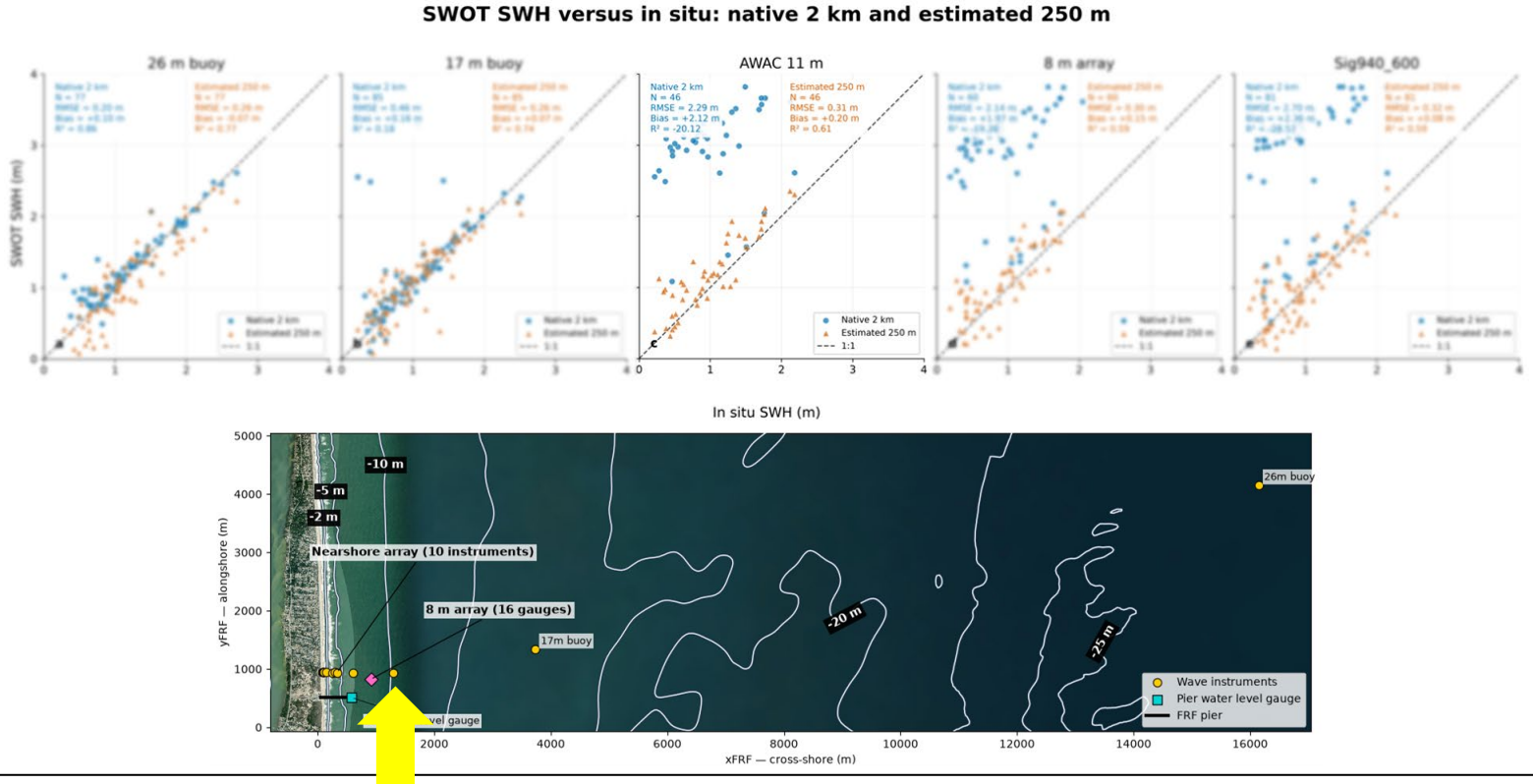
Significant wave height using LR Unsmoothed



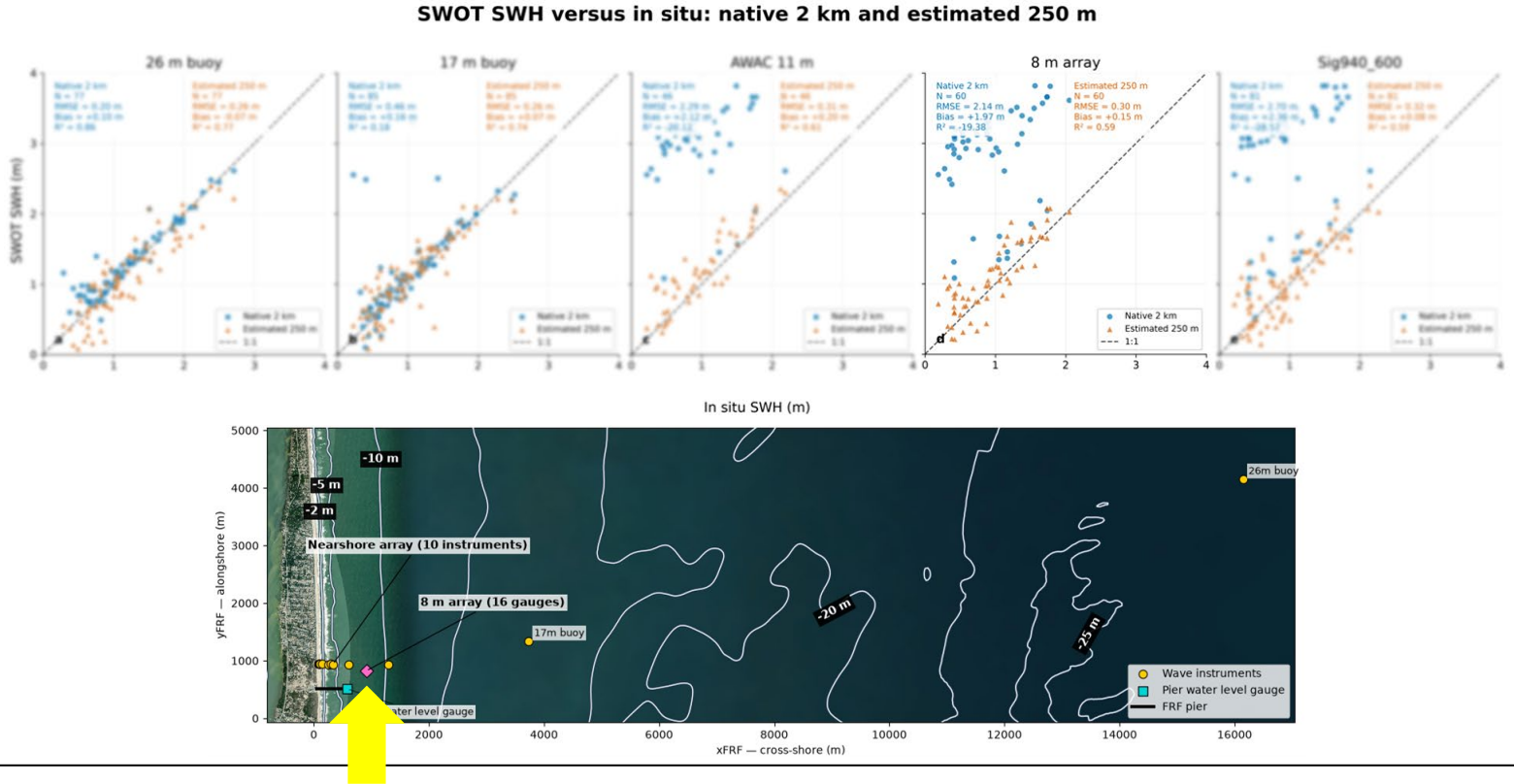
Significant wave height using LR Unsmoothed



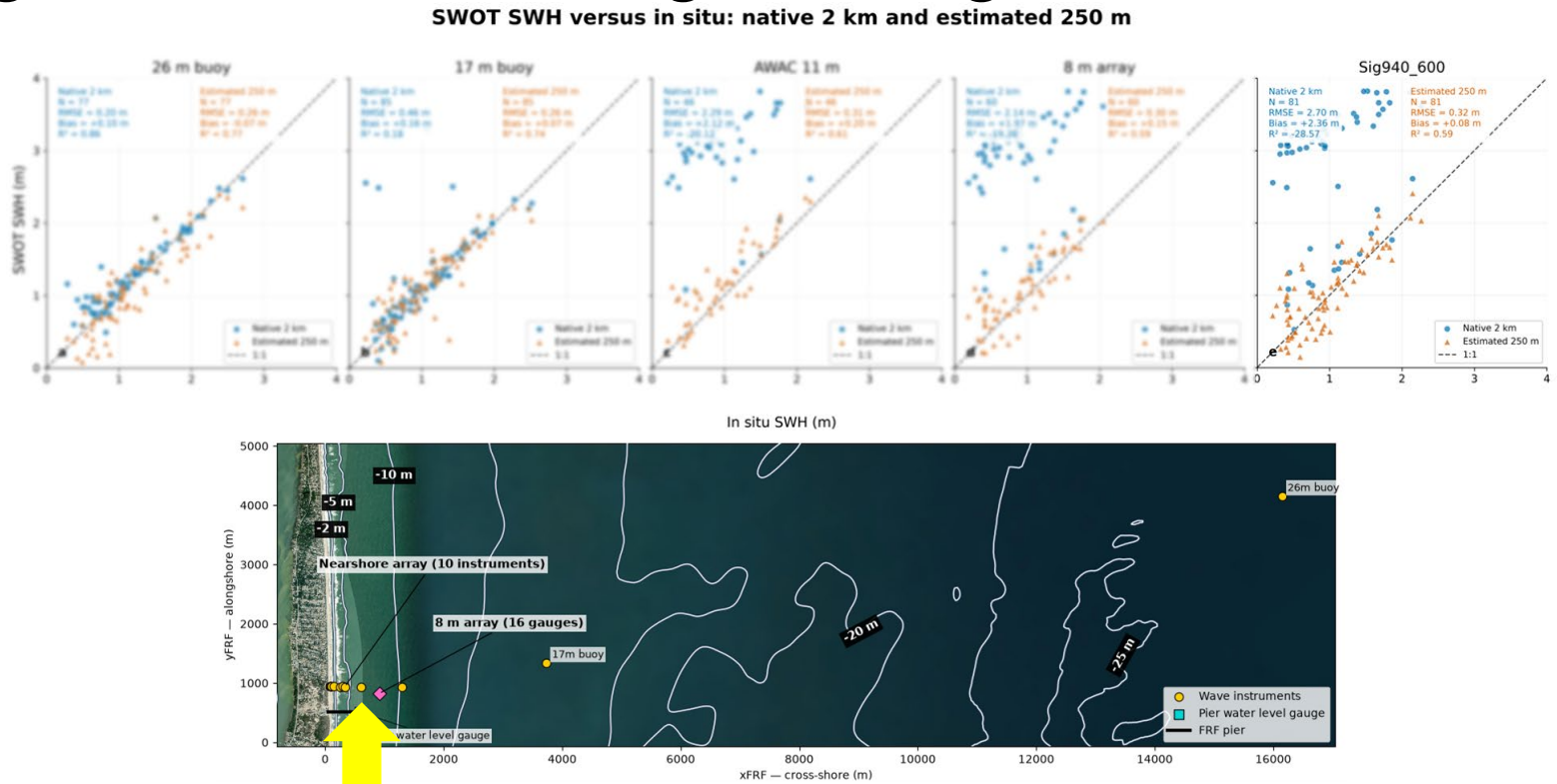
Significant wave height using LR Unsmoothed



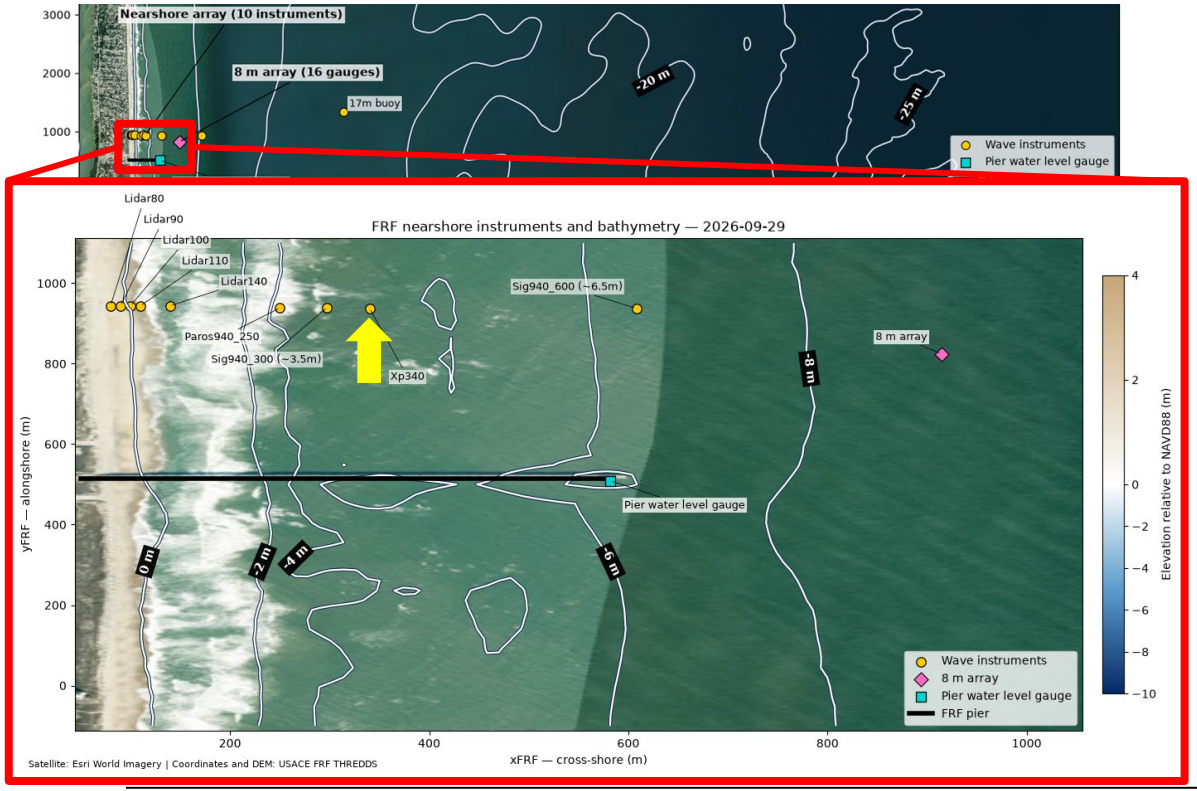
Significant wave height using LR Unsmoothed

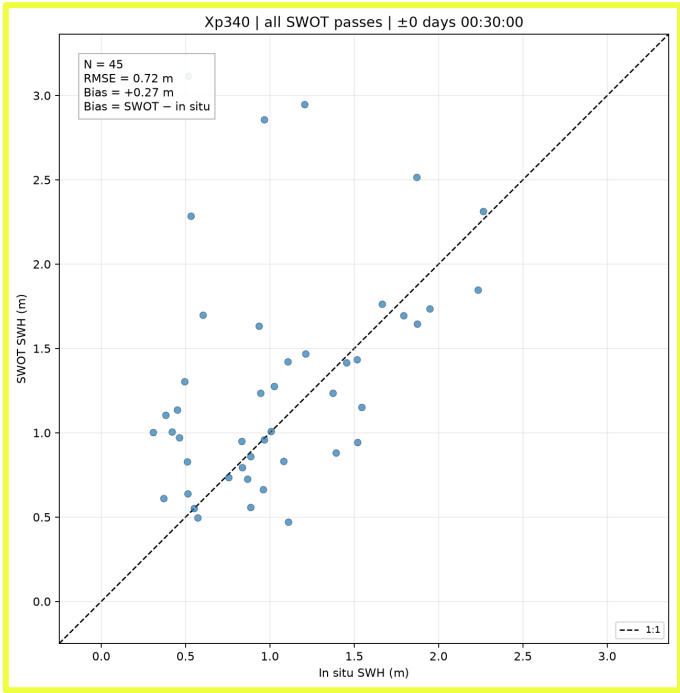


Significant wave height using LR Unsmoothed

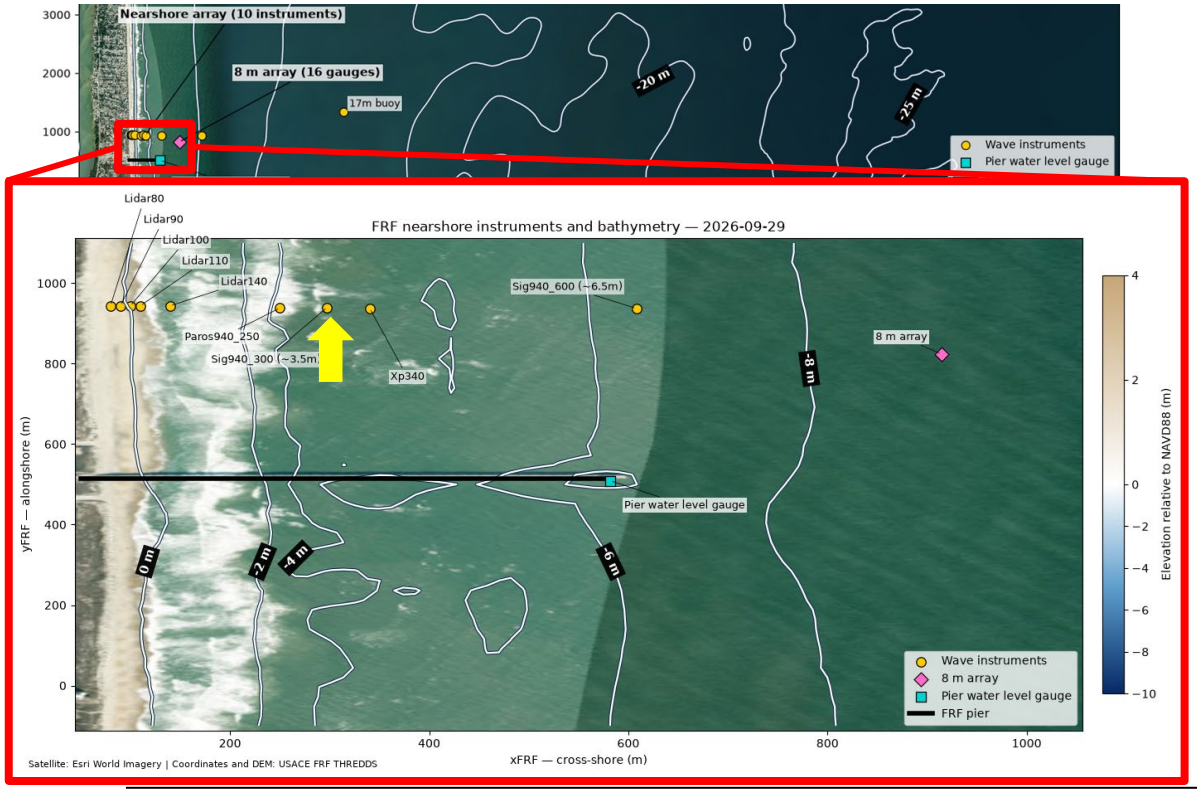


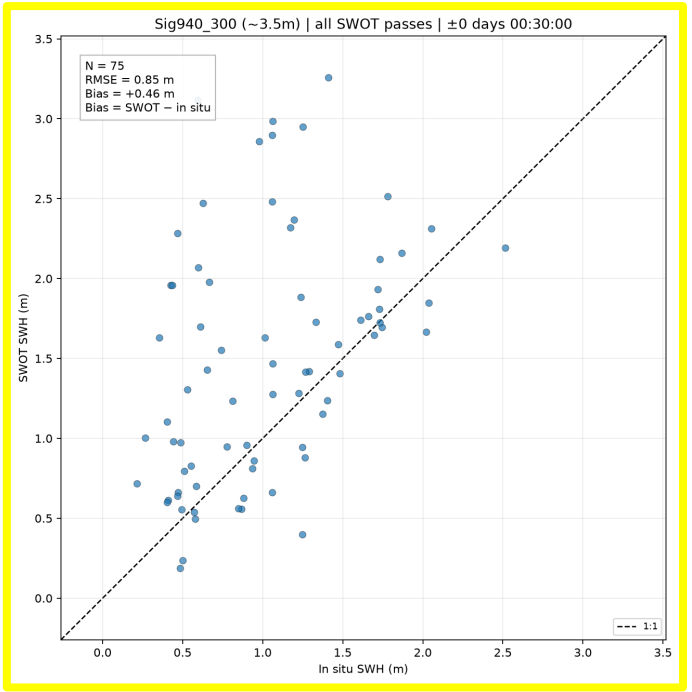
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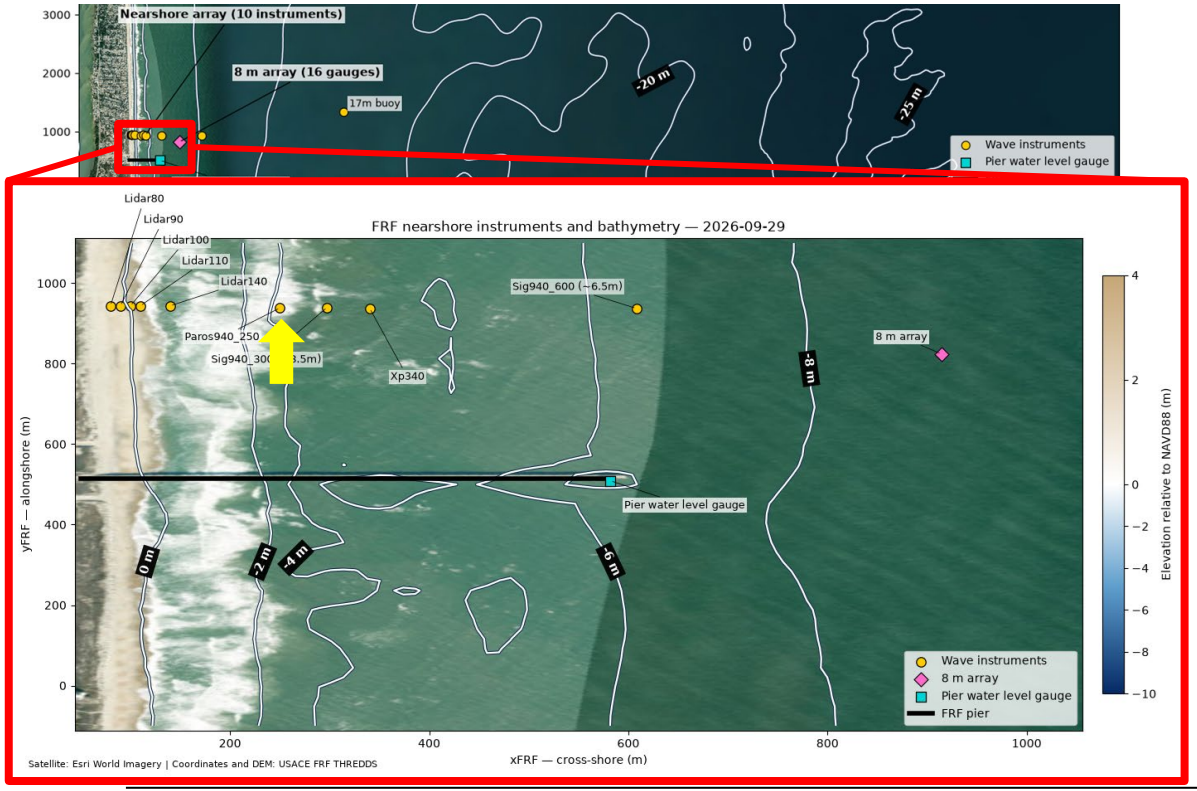


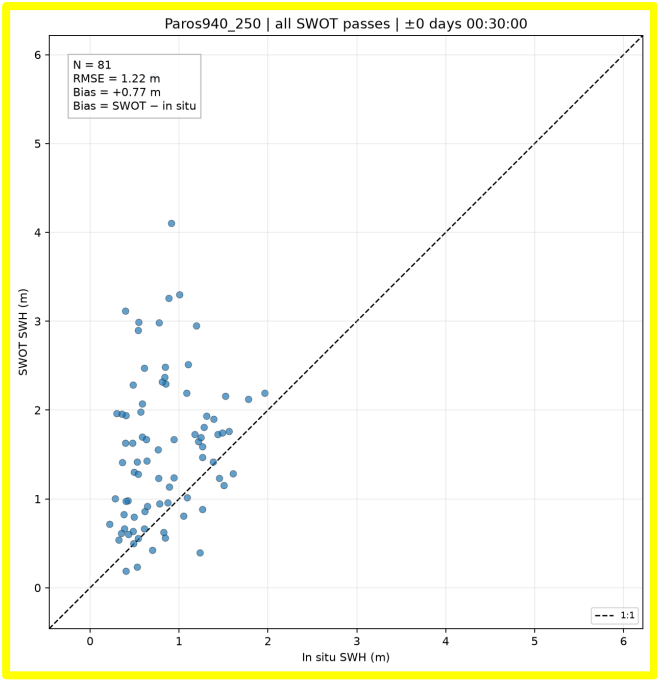
Significant wave height using LR Unsmoothed





Significant wave height using LR Unsmoothed





Discussion

WSE

- Accuracy remains good even 400 m from shore
- Tradeoffs between WSE accuracy and measurement location precision

Wave Setup

- Algorithm needs refinement
- Shows promise for SWOT derived setup and storm surge measurements

SWH

- L2 LR Unsmoothed enables SWH measurements as close as 500 m from shore

Thank you!


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Extra Slides

SWOT Processing- L3 LR SSH Unsmoothed 250 m

- L3 LR SSH 250 m, Version D
- WSE reconstruction:

$$WSE_{\mathrm{NAVD88}} = ssha_{\mathrm{filtered}} + MSS + DAC + oceanTides + \Delta_{EGM2008 \rightarrow NAVD88}$$

- Quality flag filtering:
  - Reject measurements with quality flags: extremes, mission_event, bad_swath_extremeties, not_on_sea, no_data
- Outlier threshold (per granule): ±3 robust standard deviations
- Offshore requirement: xFRF > 100 m
- 25 m exclusion zone around pier

SWOT Processing- L2 LR SSH Unsmoothed 250 m

- L2 HR Raster 100 m, Version D
- WSE reconstruction:

$$WSE_{\mathrm{NAVD88}} = ssh_karin_2 + xover_cor + SSB - geoid - SET - LT - PT + \Delta_{EGM2008 \rightarrow NAVD88}$$

- All SWOT corrections interpolated from L2 LR SSH Expert (2 km)
- Quality flag filtering:
  - Retain measurements good or suspect quality bits, reject degraded and bad
- Outlier threshold (per granule): ±3 robust standard deviations
- Offshore requirement: xFRF > 100 m
- 25 m exclusion zone around pier

SWOT Processing- L2 HR Raster 100 m

- L2 HR Raster 100 m, Version D
- WSE reconstruction:

$$WSE_{\mathrm{NAVD88}} = wse_{\mathrm{EGM2008}} + \Delta_{EGM2008 \rightarrow NAVD88}$$

- Quality flag filtering:
  - Retain WSE with wse_qual good or suspect, reject degraded and bad
- Outlier threshold (per granule): ±3 robust standard deviations
- Maximum distance to in-situ location: 50 m
- Offshore requirement: xFRF > 0 m
- 25 m exclusion zone around pier

SWOT Processing- L2 HR Water Mask PIXC

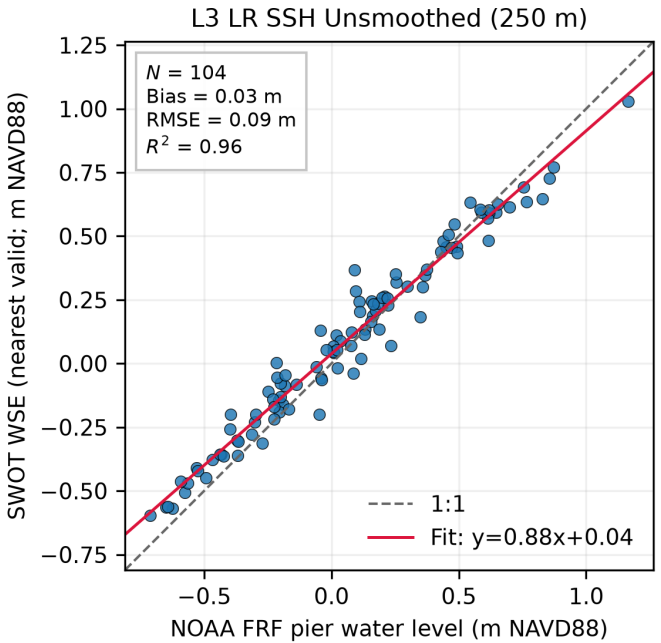
- L2 HR Water Mask Pixel Cloud, Version D
- WSE reconstruction:

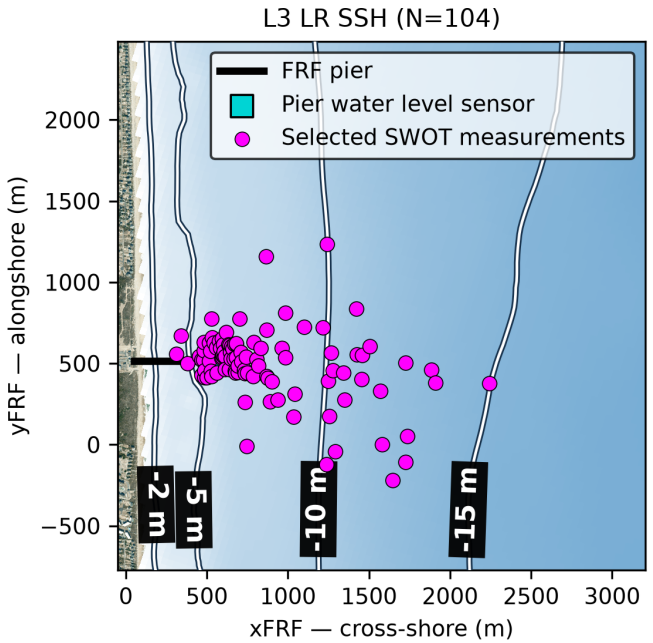
$$WSE_{\mathrm{NAVD88}} = height - geoid - SET - LT - PT + \Delta_{EGM2008 \rightarrow NAVD88}$$

- Quality flag filtering:
  - Reject pixels with sig0_qual, interferogram_qual, and geolocation_qual suspect, degraded or bad.
  - Reject phase-unwrapping flags
  - Reject land classifications 1,2,3,5,6,7 (Keep only open_water)
  - Reject bright land values of 1,2 (bright_land and bright_land_or_water)
- Outlier threshold (per granule): ±3 robust standard deviations
- Maximum distance to in-situ location: 50 m
- 25 m exclusion zone around pier
- Offshore requirement: xFRF > 0 mm

Nearshore Water Surface Elevation (WSE)

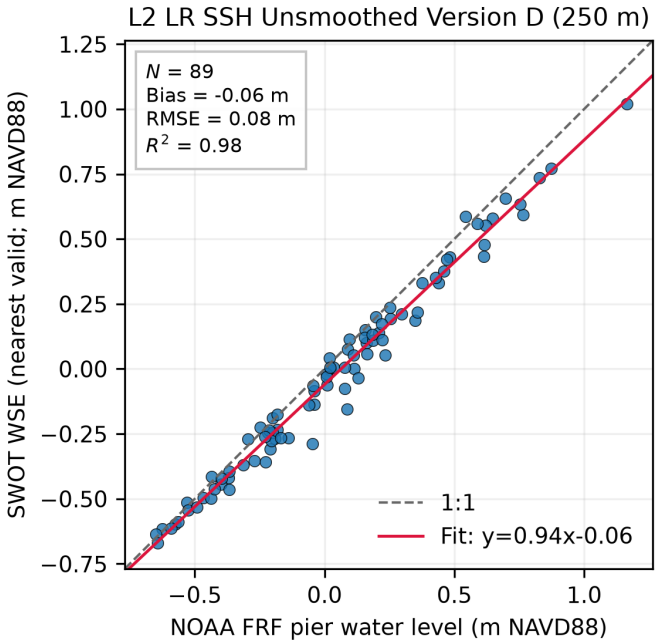
Product: L3 LR SSH Unsmoothed (250 m)

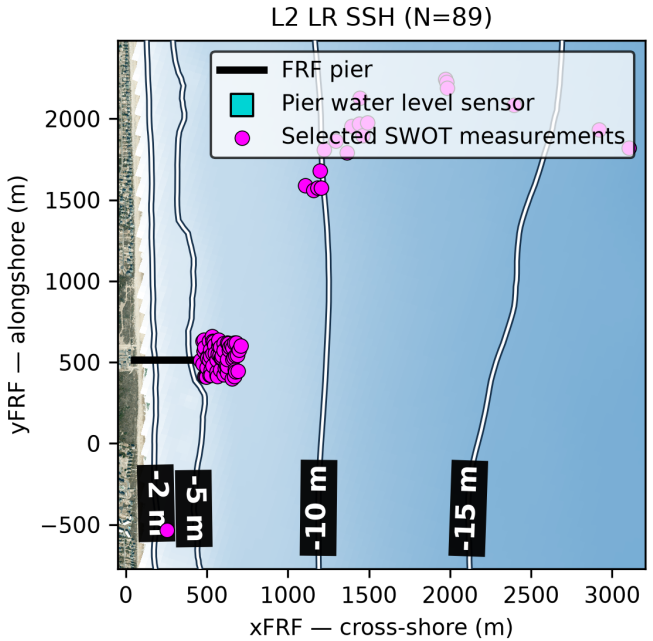




Nearshore Water Surface Elevation (WSE)

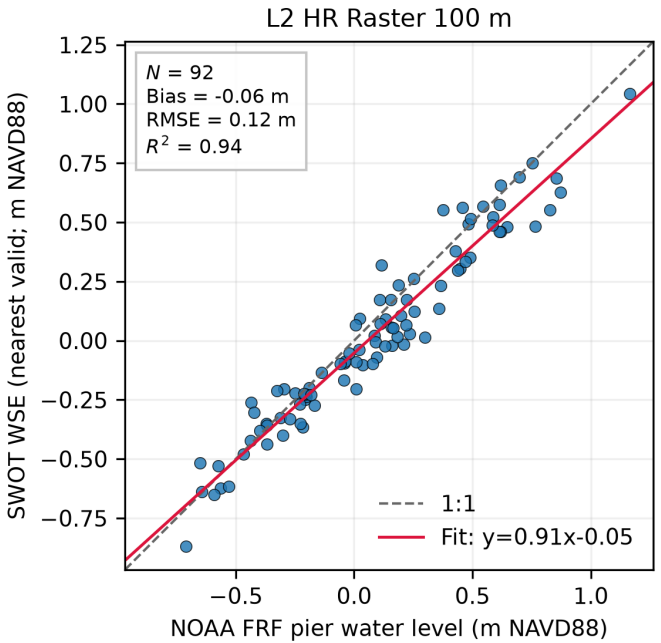
Product: L2 LR SSH Unsmoothed (250 m)

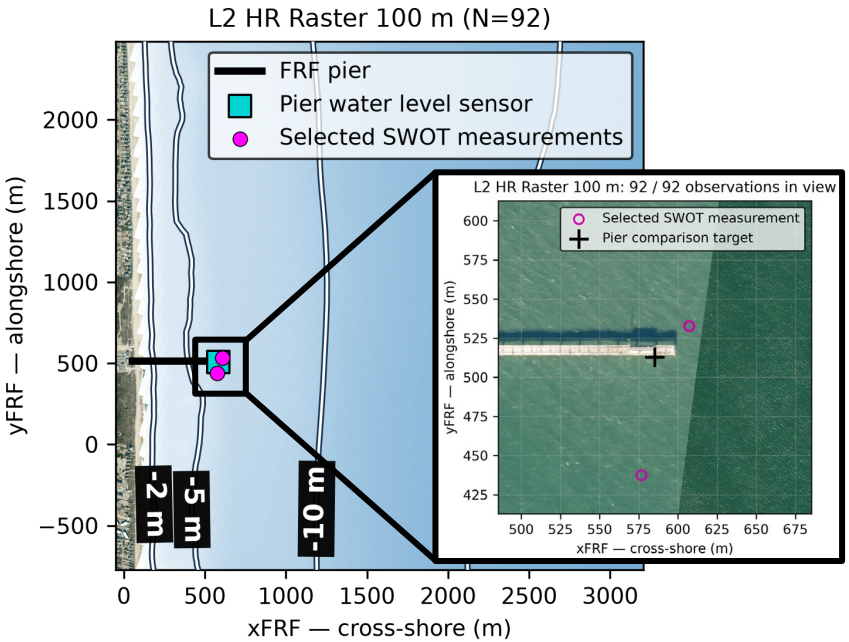




Nearshore Water Surface Elevation (WSE)

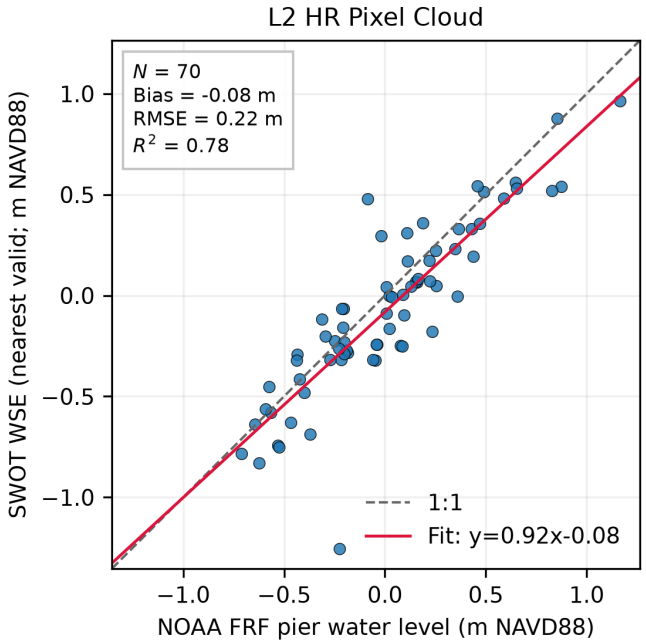
Product: L2 HR Raster (100 m)

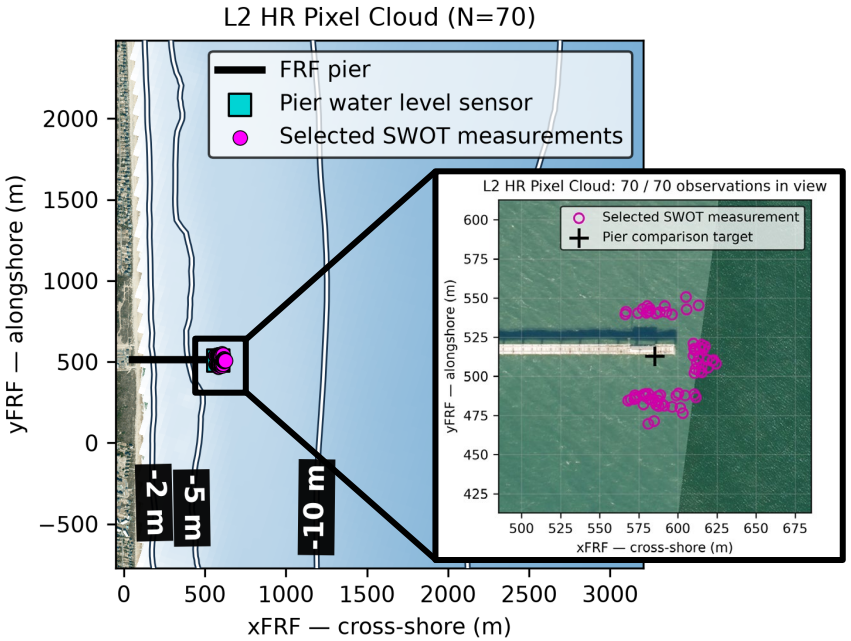




Nearshore Water Surface Elevation (WSE)

Product: L2 HR Water Mask Pixel Cloud (22 x 25/45 m)



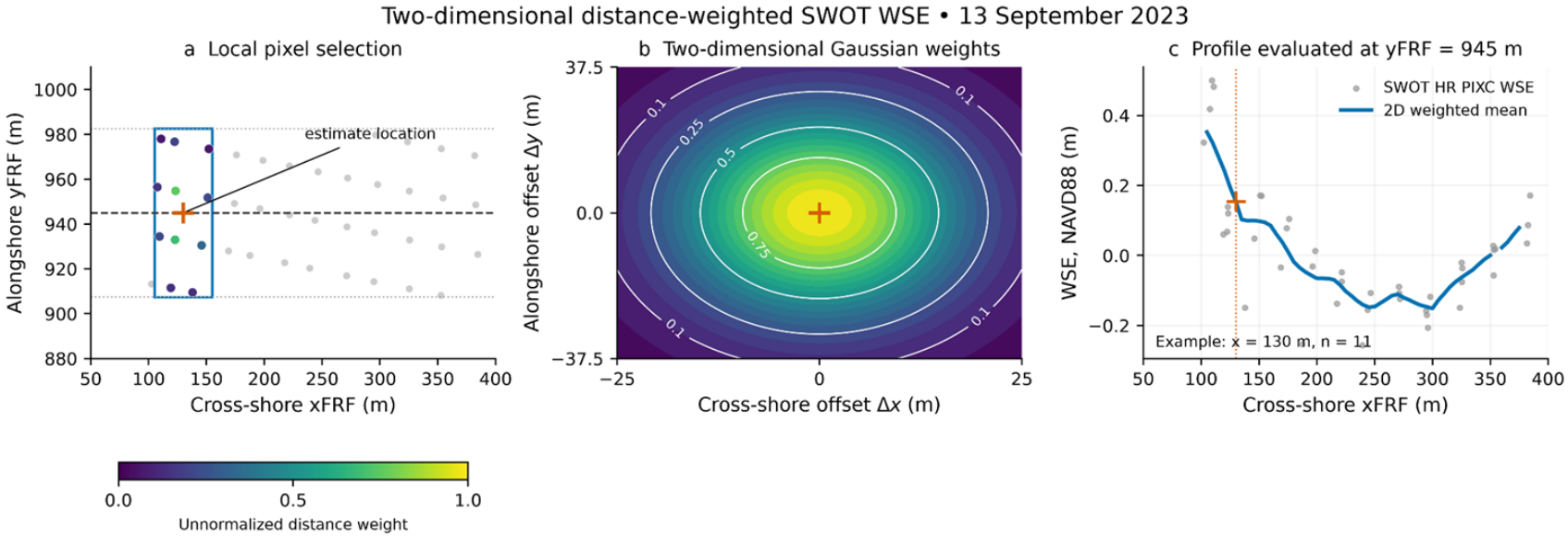


SWOT WSE Comparison Distance Summary

FRF pier — SWOT measurement distances

Product	N	Mean (m)	Median (m)	Max (m)
L3 LR SSH	104	372.2	161.9	1663.1
L2 LR SSH	89	465.6	114.1	2838.1
L2 HR Raster 100 m	92	31.1	29.6	75.8
L2 HR Pixel Cloud	70	31.0	30.2	43.3

SWOT Processing- Surfzone Water Level



$$w_i = \exp\left[-\tfrac{1}{2}\left((\Delta x_i/12.5)^2 + (\Delta y_i/18.75)^2\right)\right] \qquad \widehat{\eta}(x_0, 945) = \frac{\sum_i w_i \eta_i}{\sum_i w_i}$$

Support: $|\Delta x| \le 25$ m, $|\Delta y| \le 37.5$ m. Evaluation spacing: 5 m. At least 3 finite pixels; no cross-shore extrapolation.

OBSERVED DATA • SWOT near lidar 2023-09-13 13:59:55 UTC • 42 quality-filtered pixels in yFRF = 945 ± 37.5 m.

SWOT Wave Setup

- SWOT transect: Averaged pixels within ±75 m of the y = 940 m transect
- Gridding: Averaged onto a 25 m alongshore grid
- Processing: Despiked, then smoothed with a 125 m moving mean
- Reference water level: Removed the mean WSE over x = 3000–4000 m
- Setup: Defined as the maximum residual WSE over x = 50–250 m

FRF 8 m array detail view

